

Project Proposal

C-DE211: Data Analysis | Spring 2026

Course	C-DE211: Data Analysis
Submission Date	21st April 2026
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1. Dataset

Name	Egyptian Real Estate Listings
Source	Kaggle — https://www.kaggle.com/datasets/hassankhaled21/egyptian-real-estate-listings
Size	19,925 rows × 12 columns CSV
Data Type	Numerical (price, area, bedrooms) & Categorical (location, property type)
Origin	Real-world data scraped from Egyptian real estate platforms — not AI-generated

2. Problem Statement

The Egyptian real estate market has experienced significant price changes in recent years. Understanding what drives property prices — whether it is the size, number of rooms, or location — is valuable for buyers and sellers alike. This project uses real listing data to identify the key factors influencing residential property prices across Egypt.

3. Research Hypothesis

H0: There is no significant relationship between property features (area, bedrooms, location) and price.

H1: Properties with larger area, more bedrooms, and premium locations have significantly higher prices in the Egyptian real estate market.

4. Project Workflow

Step	Phase	Description
1	Data Collection	Load dataset from Kaggle (19,925 rows × 12 columns)
2	Data Cleaning	Handle missing values, fix data types, remove duplicates
3	EDA	Explore distributions, correlations, and patterns
4	Visualization	Charts for price vs. area, location, bedrooms
5	Hypothesis Testing	Statistical tests to accept or reject H0
6	ML Model	Regression model to predict property prices